

## WINDOWS FORENSICS | PROJECT: ANALYZER

UNIT: RW-University-II

Name: YAHAYA MEDDY

CODE: s10

### 1. Project Overview

The Forensic Analysis Automation Tool is a comprehensive Bash-based script designed to automate the analysis of hard disk images and memory dumps for forensic investigations. This tool integrates multiple powerful utilities like **bulk\_extractor**, **strings**, and **Volatility** to streamline data extraction, keyword searches, memory profiling, and more.

### 2. Objectives

The primary objectives of the tool are:

1. Automate HDD and memory analysis for forensic purposes.
2. Provide a user-friendly, interactive approach to forensic investigations.
3. Simplify the installation and validation of required tools.
4. Ensure compatibility and proper handling of provided files for analysis.
5. Generate detailed reports of the extracted data for further review.

### 3. Features and Functionalities

#### 1. Root Privilege Check

Ensures the script is executed with root privileges to perform forensic tasks effectively.

#### 2. Tool Installation and Validation

Verifies the presence of required tools (**bulk\_extractor**, **strings**, **Volatility**, and **zip**). Automatically installs missing tools and prompts the user to ensure Volatility is accessible.

### 3. HDD Analysis (Bulk Extractor)

- Runs **bulk\_extractor** on the user-provided file.
- Extracts and identifies network traffic data in **pcap** format, if available.
- Provides size and location details of identified **.pcap** files.

### 4. Keyword Search (Strings)

- Allows users to specify a file and enter keywords for searching.
- Searches for human-readable strings in the file and outputs results into a report.

### 5. Memory Analysis (Volatility)

- Identifies if the memory file can be analyzed with Volatility.
- Detects the memory profile automatically.
- Extracts **running processes**, **network connections**, and **registry information** into separate files.

### 6. Result Packaging

- Consolidates all analysis outputs into a timestamped directory.
- Compresses the results directory into a zip file for easy sharing and storage.

## 4. User Workflow

### 1. Setup and Validation

- The script checks for root access and installs required tools.
- Ensures all files provided by the user exist and are valid.

### 2. Input Collection

- Users provide two files:
  - A file for **bulk\_extractor** and memory analysis.
  - A file for keyword searches.

### 3. Analysis Execution

The script sequentially performs:

1. HDD analysis with **bulk\_extractor**.
2. Keyword searches with **strings**.
3. Memory analysis with **Volatility**.

### 4. Report Generation

- Outputs general statistics, analysis files, and other results into a structured directory.
- Compresses the directory into a zip file.

## 5. Sample Output

Step 1: Checking Required Tools and user inputs of the files that need to be analysed.

```
(kali㉿kali)-[~/Desktop/cyberium/project]
$ sudo ./Analyzer.sh
[sudo] password for kali:

[START] Forensic Analyzer

[STEP 1: Checking Required Tools]

[INFO] Volatility (vol) is available.
[INFO] bulk_extractor is already installed.
[INFO] strings is already installed.
[INFO] zip is already installed.

[INFO] Tool check complete.

[User Input]

Enter the file to use for bulk_extractor and memory analysis:
memdump.mem
Enter the file to use for strings analysis:
carvme
```

Step 2: HDD or memory analysis using **bulk\_extractor** and find if the results contains a **pcap** file.

```
[STEP 2: HDD Analysis]
```

```
[INFO] Running bulk_extractor on memdump.mem ...  
bulk_extractor version: 2.1.1  
Input file: "memdump.mem"  
Output directory: "bulk_extractor_output"  
Disk Size: 536870912  
Scanners: aes base64 elf evtx exif facebook find gzip httplogs json kml_  
ch zip accts email gps  
Threads: 4  
going multi-threaded... ( 4 )  
bulk_extractor      Tue Jan 21 00:24:36 2025
```

```
Average time each consumer spent waiting for data from producer: 0:00:00 (0.00 seconds)  
*** More time spent waiting for workers. You need a faster CPU or more cores for improved performance.  
Total email features found: 123  
[INFO] PCAP file found: bulk_extractor_output/packets.pcap (Size: 104K)  
[INFO] HDD Analysis complete.
```

Step 3: Allows a user to input the keywords they want to search in a specific file.

```
[STEP 3: Keyword Search]
```

```
Enter keywords to search (separated by spaces):
```

```
email username password credit bank
```

```
[INFO] Searching for keywords in carvme...
```

```
[INFO] Results saved to strings_results.txt.
```

```
[INFO] Keyword Search complete.
```

Step 4: Memory analysis using volatility

```
[STEP 4: Memory Analysis]
```

```
[INFO] Checking if memdump.mem can be analyzed with Volatility...
```

```
Volatility Foundation Volatility Framework 2.5
```

```
INFO : volatility.debug : Determining profile based on KDBG search...
```

```
Volatility Foundation Volatility Framework 2.5
```

```
INFO : volatility.debug : Determining profile based on KDBG search...
```

```
[INFO] Detected memory profile: WinXPSP3x86
```

```
[INFO] Listing processes...
```

```
Volatility Foundation Volatility Framework 2.5
```

```
[INFO] Listing network connections...
```

```
Volatility Foundation Volatility Framework 2.5
```

```
[INFO] Extracting registry information...
```

```
Volatility Foundation Volatility Framework 2.5
```

```
[INFO] Memory Analysis complete.
```

Step 5: Moving the all the results in a single directory and zip the directory.

[STEP 5: Packaging Results]

```
[INFO] Zipping results ...  
  adding: analysis_results_20250121_002810/ (stored 0%)  
  adding: analysis_results_20250121_002810/registry.txt (deflated 70%)  
  adding: analysis_results_20250121_002810/strings_results.txt (deflated 38%)  
  adding: analysis_results_20250121_002810/processes.txt (deflated 55%)  
  adding: analysis_results_20250121_002810/packets.pcap (deflated 9%)  
  adding: analysis_results_20250121_002810/connections.txt (deflated 77%)  
  
[INFO] Analysis complete. Results saved in 'analysis_results_20250121_002810.zip'.
```

Created files:

```
(kali㉿kali)-[~/Desktop/cyberium/project]  
$ ls analysis_results_20250120_140932.zip  
analysis_results_20250120_140932.zip  
  
(kali㉿kali)-[~/Desktop/cyberium/project]  
$ ls analysis_results_20250120_140932  
connections.txt  packets.pcap  processes.txt  registry.txt  strings_results.txt
```

## Connections.txt file content

```
(kali㉿kali)-[~/Desktop/cyberium/project]
$ cat analysis_results_20250120_140932/connections.txt
```

| Offset(V)  | PID  | Port | Proto | Protocol | Address        | Create Time                  |
|------------|------|------|-------|----------|----------------|------------------------------|
| 0x81ddb780 | 664  | 500  | 17    | UDP      | 0.0.0.0        | 2012-07-22 02:42:53 UTC+0000 |
| 0x82240d08 | 1484 | 1038 | 6     | TCP      | 0.0.0.0        | 2012-07-22 02:44:45 UTC+0000 |
| 0x81dd7618 | 1220 | 1900 | 17    | UDP      | 172.16.112.128 | 2012-07-22 02:43:01 UTC+0000 |
| 0x82125610 | 788  | 1028 | 6     | TCP      | 127.0.0.1      | 2012-07-22 02:43:01 UTC+0000 |
| 0x8219cc08 | 4    | 445  | 6     | TCP      | 0.0.0.0        | 2012-07-22 02:42:31 UTC+0000 |
| 0x81ec23b0 | 908  | 135  | 6     | TCP      | 0.0.0.0        | 2012-07-22 02:42:33 UTC+0000 |
| 0x82276878 | 4    | 139  | 6     | TCP      | 172.16.112.128 | 2012-07-22 02:42:38 UTC+0000 |
| 0x82277460 | 4    | 137  | 17    | UDP      | 172.16.112.128 | 2012-07-22 02:42:38 UTC+0000 |
| 0x81e76620 | 1004 | 123  | 17    | UDP      | 127.0.0.1      | 2012-07-22 02:43:01 UTC+0000 |
| 0x82172808 | 664  | 0    | 255   | Reserved | 0.0.0.0        | 2012-07-22 02:42:53 UTC+0000 |
| 0x81e3f460 | 4    | 138  | 17    | UDP      | 172.16.112.128 | 2012-07-22 02:42:38 UTC+0000 |
| 0x821f0630 | 1004 | 123  | 17    | UDP      | 172.16.112.128 | 2012-07-22 02:43:01 UTC+0000 |
| 0x822cd2b0 | 1220 | 1900 | 17    | UDP      | 127.0.0.1      | 2012-07-22 02:43:01 UTC+0000 |
| 0x82172c50 | 664  | 4500 | 17    | UDP      | 0.0.0.0        | 2012-07-22 02:42:53 UTC+0000 |
| 0x821f0d00 | 4    | 445  | 17    | UDP      | 0.0.0.0        | 2012-07-22 02:42:31 UTC+0000 |

## Processes.txt file content

```
(kali㉿kali)-[~/Desktop/cyberium/project]
$ cat analysis_results_20250120_140932/processes.txt
```

| Name          |
|---------------|
| System        |
| smss.exe      |
| csrss.exe     |
| winlogon.exe  |
| services.exe  |
| lsass.exe     |
| svchost.exe   |
| svchost.exe   |
| svchost.exe   |
| svchost.exe   |
| svchost.exe   |
| explorer.exe  |
| spoolsv.exe   |
| reader_sl.exe |
| alg.exe       |
| wuaclt.exe    |
| wuaclt.exe    |

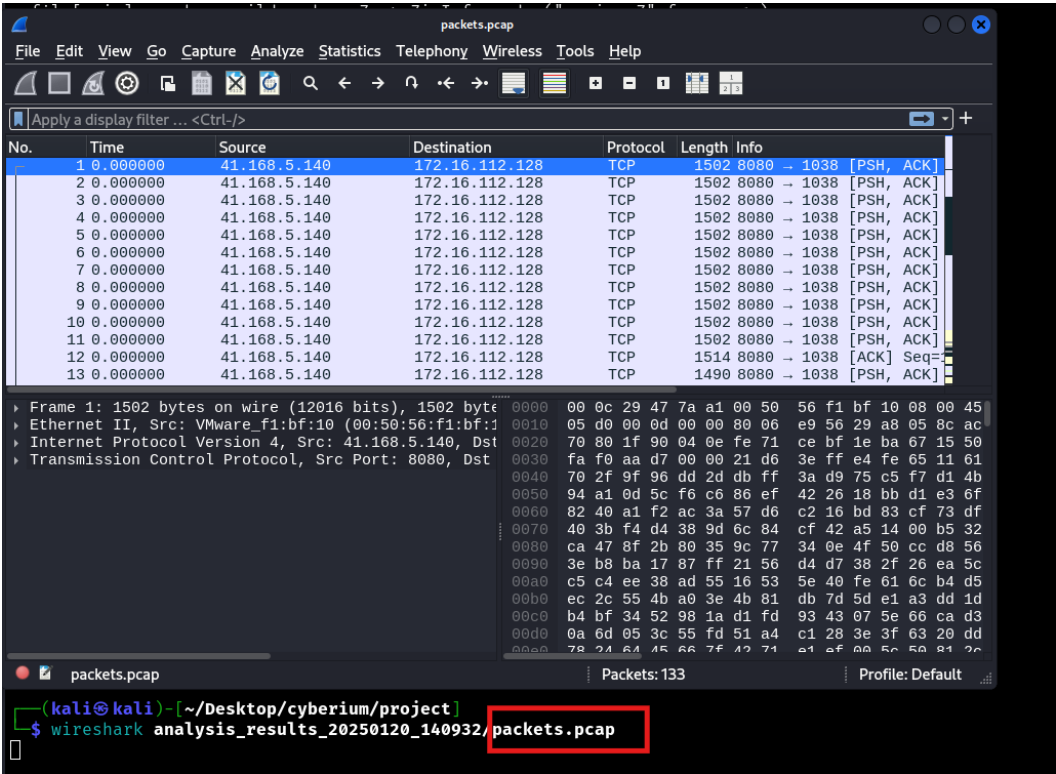
## Registry.txt file content

```
(kali@kali) - [~/Desktop/cyberium/project]
$ cat analysis_results_20250120_140932/registry.txt
Virtual    Physical    Name
0xe18e5b60 0x093f8b60 \Device\HarddiskVolume1\Documents and Settings\Robert\Local Settings\Application Data\Microsoft\Windows\UsrClass.dat
0xe1a19b60 0x0a5a9b60 \Device\HarddiskVolume1\Documents and Settings\Robert\NTUSER.DAT
0xe18398d0 0x08a838d0 \Device\HarddiskVolume1\Documents and Settings\LocalService\Local Settings\Application Data\Microsoft\Windows\UsrClass.dat
0xe18614d0 0x08e624d0 \Device\HarddiskVolume1\Documents and Settings\LocalService\NTUSER.DAT
0xe183bb60 0x08e2db60 \Device\HarddiskVolume1\Documents and Settings\NetworkService\Local Settings\Application Data\Microsoft\Windows\UsrClass.dat
0xe17f2b60 0x08519b60 \Device\HarddiskVolume1\Documents and Settings\NetworkService\NTUSER.DAT
0xe1570510 0x07669510 \Device\HarddiskVolume1\WINDOWS\system32\config\software
0xe1571008 0x0777f008 \Device\HarddiskVolume1\WINDOWS\system32\config\default
0xe15709b8 0x076699b8 \Device\HarddiskVolume1\WINDOWS\system32\config\SECURITY
0xe15719e8 0x0777f9e8 \Device\HarddiskVolume1\WINDOWS\system32\config\SAM
0xe13ba008 0x02e4b008 [no name]
0xe1035b60 0x02ac3b60 \Device\HarddiskVolume1\WINDOWS\system32\config\system
0xe102e008 0x02a7d008 [no name]
```

## Strings\_results.txt file content

```
(kali@kali) - [~/Desktop/cyberium/project]
$ cat analysis_results_20250121_002810/strings_results.txt
Username : jesseselkirk@gmail.com
Password : K1L2Q3456JnK$
Vorname: Justin
Vorname: Torben
Vorname: Mateus
Vorname: Kevin
Vorname: Stefanie
Vorname: Claudia
Bank Account Number : 4016286759086
BIN/IIN Info : MASTERCARD - CREDIT - BUSINESS - CITIBANK, N.A.
BIN/IIN Info : MASTERCARD - CREDIT - BUSINESS - CITIBANK, N.A.
Hacked Credit Card List Details 2021
```

Packets.pcap file content



The image shows a Wireshark window titled 'packets.pcap'. The packet list pane on the left shows 13 packets. The packet details pane on the right shows the selected packet (No. 1) with details for Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane at the bottom shows the raw data of the selected packet. The status bar at the bottom indicates 'Packets: 133' and 'Profile: Default'.

| No. | Time     | Source       | Destination    | Protocol | Length | Info                   |
|-----|----------|--------------|----------------|----------|--------|------------------------|
| 1   | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 2   | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 3   | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 4   | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 5   | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 6   | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 7   | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 8   | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 9   | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 10  | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 11  | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1502   | 8080 → 1038 [PSH, ACK] |
| 12  | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1514   | 8080 → 1038 [ACK] Seq= |
| 13  | 0.000000 | 41.168.5.140 | 172.16.112.128 | TCP      | 1490   | 8080 → 1038 [PSH, ACK] |

Frame 1: 1502 bytes on wire (12016 bits), 1502 byte  
Ethernet II, Src: VMware\_f1:bf:10 (00:50:56:f1:bf:10), Dst: 02:00:00:00:00:00 (00:00:00:00:00:00)  
Internet Protocol Version 4, Src: 41.168.5.140, Dst: 172.16.112.128  
Transmission Control Protocol, Src Port: 8080, Dst Port: 1038

packets.pcap | Packets: 133 | Profile: Default

```
(kali@kali) - [~/Desktop/cyberium/project]
$ wireshark analysis_results_20250120_140932.packets.pcap
```

Zip file content

```
(kali@kali) - [~/Desktop/cyberium/project]
$ unzip -l analysis_results_20250120_140932.zip
Archive:  analysis_results_20250120_140932.zip
Length      Date       Time       Name
-----
0           2025-01-20 14:09  analysis_results_20250120_140932/
1211        2025-01-20 14:09  analysis_results_20250120_140932/registry.txt
258         2025-01-20 14:08  analysis_results_20250120_140932/strings_results.txt
219         2025-01-20 14:09  analysis_results_20250120_140932/processes.txt
103629      2025-01-20 14:06  analysis_results_20250120_140932/packets.pcap
1581        2025-01-20 14:09  analysis_results_20250120_140932/connections.txt
-----
106898                                6 files

(kali@kali) - [~/Desktop/cyberium/project]
$
```

## Conclusion

The Forensic Analysis Automation Tool simplifies digital forensic investigations by automating disk and memory analysis processes. It ensures consistency, efficiency, and user-friendliness, making it an essential asset for forensic practitioners. This tool demonstrates how powerful scripting can streamline complex tasks in a forensically sound manner.